

Market Basket Analysis



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Agenda

- Introduction
- Analytic Terms
- Tool Creation Goals
- Benefits and Constraints
- Conclusion
- Q&A

Introduction

- Affinity Analysis → Market Basket Analysis (MBA)
 - Data mining technique: conditional probability (Bayes' Theorem)
 - Discover the co-occurrence of relationships among products (e.g. beers and diapers?)
- Widely adopted by different industries for various purposes
 - Retail & E-Commerce: cross-sell and upsell, bundle/discount, shelf/page display
 - Finance: credit card purchase pattern and fraudulent prevention
- Easy to understand and implement
 - Gather data with a defined basket
 - e.g.// basket defined by customers, by transactions...etc

Analytic Terms - I

- **Occurrence / Co-occurrence:** Number of events of an item (or set) being purchased among the group (customers buying ≥ 2 different products)
 - e.g.// Let there be...
 - **100** Customers in the group, among them,
 - Bread is purchased by **50** customers,
 - Butter is purchased by **30** customers,
 - Bread and Butter are both purchased by the same 100 customers occurred **20** times.
- **Support:** Frequency (ratio) of occurrence in the group
 - e.g.//
 - $\text{Support}(\text{Bread}): 50 / 100 = 0.5$
 - $\text{Support}(\text{Butter}): 30 / 100 = 0.3$
 - $\text{Support}(\text{Co-occurrence}): 20 / 100 = 0.2$

Analytic Terms - II

- **Confidence:** The probability of both items being purchased based on one item being purchased

- e.g.//

$$\begin{aligned}\text{Confidence}(\text{Bread}) &= \text{Support}(\text{Co-occur}) / \text{Support}(\text{Bread}) \\ &= 0.2 / 0.5 = 0.4\end{aligned}$$

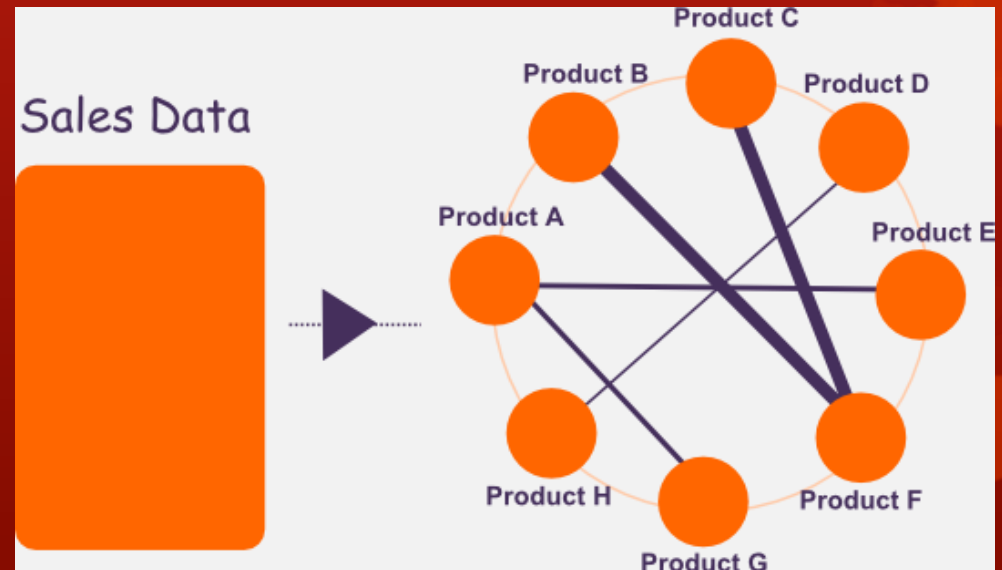
$$\begin{aligned}\text{Confidence}(\text{Butter}) &= \text{Support}(\text{Co-occur}) / \text{Support}(\text{Butter}) \\ &= 0.2 / 0.3 = 0.6667\end{aligned}$$

- **Lift:** Key indicator, the likelihood of two products going together
 - Only meaningful with values greater than “1”, the higher the better
 - However, no market standard on how high the value should be
 - Value is determined relatively among all outputs
 - e.g.// $\text{Lift}(\text{Bread \& Butter})$
$$\begin{aligned}&= \text{Support}(\text{Co-occur}) / (\text{Support}(\text{Bread}) \times \text{Support}(\text{Butter})) \\ &= 0.2 / 0.15 = 1.33\end{aligned}$$

Cross-Selling Tool Creation Goals

The tool is designed to meet the following criteria:

- **Goal 1:** To satisfy cross-selling strategy
- **Goal 2:** Easy to implement and to use among business units
- **Goal 3:** Easy to maintain and to update (automation)



Benefits

Why MBA model?

- Accurate and easy to read
- Automation is possible
 - “All products to be analyzed under 10 minutes” versus “correlation analysis, comparing 3-4 products per day”
- The balance between data cleanness versus time cost
 - Different levels of cleanness still show similar patterns
- Additional information on cross-selling potentials
 - Product popularity (popular products versus unpopular products)
 - Compare by years, seasons, stores ... etc.

Constraints

Current limitations on using this model:

- Products with similar “product format” are still being compared
 - e.g. 1L milk vs 2L milk
 - Contingency plan: Use filtering feature in Excel to screen products
- Basic MBA by Excel
 - We can only compare items on 1 to 1 relationships (A and B relationship)
 - Advanced MBA ($A + B + C \rightarrow D$) requires more advanced software services
- Data cleaning and model modification
 - Power Query training is required
 - Raw data need to be cleaned to reduce model running time
 - Procedures will need to be modified or re-created if there is change in analysis goals (e.g. changes in basket / product hierarchy...etc)

Conclusion - SMART

- **S**pecific: can be measured under different product hierarchies
- **M**easurable: indicators (Support, Confidence, Lift) support decisions
- **A**chievable: cost-effective if there is no budget to acquire external software license
- **R**elevant: match cross-selling strategy
- **T**imeliness: automation with Power Query, easy to learn, fast result

The End

